SOVEREIGN INDUSTRIAL ENGINEERING AUDIT

EQUIPMENT INTEGRITY & INSPECTION REPORT

DOCUMENT NO: MRPL-INSP-2025-E204

INSPECTION DATE: October 14, 2025

FACILITY: MRPL Refining Complex - CDU-1

NEXT INSPECTION: October 2026 (Due to wear)

1.0 GENERAL EQUIPMENT SPECIFICATION

Equipment Tag:	E-204 (Crude Pre-flash Column Overhead Condenser)
Equipment Type:	TEMA Class R - Shell and Tube Heat Exchanger (AES Type)
Design Code:	ASME Section VIII Division 1 (2019 Edition) / TEMA Class R
Inspection Standard:	API 510 (Pressure Vessel Inspection) & API 579-1 / ASME FFS-1
Design Pressure:	Shell Side: 15.5 bar (g) | Tube Side: 18.0 bar (g)
Design Temperature:	Shell Side: 185 deg C | Tube Side: 110 deg C
Operating Fluid:	Shell: Hydrocarbon vapor + H2S | Tube: Cooling Water
Shell Material:	SA-516 Grade 70 Carbon Steel (Normalized)
Nominal Shell Thickness:	10.0 mm
Corrosion Allowance (CA):	3.0 mm
Minimum Required (t_min):	7.8 mm (calculated per ASME Section VIII UG-27)

2.0 EXECUTIVE SUMMARY & CRITICAL ALERT

3.0 ULTRASONIC THICKNESS MEASUREMENT (UTM) RESULTS

Component Location	Nominal (mm)	Min Req t_min (mm)	Measured (mm)	Status
Shell Course 1 (Inlet Nozzle N3)	10.0	7.8	8.2	WARNING
Shell Course 2 (Center Ring)	10.0	7.8	9.1	NORMAL
Shell Course 3 (Outlet)	10.0	7.8	9.4	NORMAL
Channel Head (Shell Side)	12.0	9.2	11.4	NORMAL
Floating Head Cover	14.0	10.5	13.2	NORMAL
Stationary Tubesheet	25.0	20.0	24.6	NORMAL
Nozzle N1 (Crude Inlet 10-inch)	9.27	6.5	8.6	NORMAL
Nozzle N3 (Vapor Return 8-inch)	8.18	5.8	6.4	WARNING

4.0 ENGINEERING CORROSION & REMAINING LIFE CALCULATIONS

5.0 RISK ASSESSMENT & FITNESS-FOR-SERVICE EVALUATION

6.0 MANDATORY ENGINEERING RECOMMENDATIONS

1. Immediate Operating Pressure Review: Re-rate Maximum Allowable Working Pressure (MAWP) from 15.5 bar to 14.0 bar pending Level 2 assessment.

2. Inspection Interval Reduction: Reduce UTM re-inspection frequency from 60 months (5 years) to 12 months (1 year). Schedule next scan for October 2026.

3. Mitigation: Install an internal stainless steel (Inconel 625 or 316L) impingement plate / weld overlay beneath Nozzle N3 during the Q1 2026 turnaround.

4. Online Corrosion Monitoring: Deploy electrical resistance (ER) corrosion probes on the overhead vapor line upstream of E-204.

5. Engineering Review Sign-off: Certified pressure vessel engineer (API 510) formal sign-off required prior to plant re-commissioning.

Lead Inspection Engineer (API 510 Cert #48291)

Chief Technical Integrity Officer, MRPL